



README

Latest Update (6/9):

- Play the completed logistics games:

- [Eggonomy](#)
- [Bio-Logistics](#)
- [War Commander](#)
- [Routebound](#)

- Attendance for June 10th and 11th is only required if you have not met the 80% minimum in that course.
- All Completes and Efforts from your Assignments and Activities are due by Sunday, June 14th!
- [You may create one portfolio site per course of mine to replace your lowest quiz grade in that course.](#)
- [Quiz 3 questions.](#)
- [Rubric](#) updated for clarity!
- LOG & TRADE: Check the [grade viewer](#) for your grade. A max of 10 (Completes + Efforts) is required for the 80% (see RUBRIC).
- Leave a review for our hard worker:
<https://jonycap.github.io/Snuggly-web-spot/hub.html>

How to Use This File:

This document will be your guide to all classes and schedules of weekly activities. It will be updated frequently, SO CHECK OFTEN! **Note: You can check the Latest Update section above or the document details to see changes made in the version history from the Clock Symbol in the upper right of this window (desktop view).**

- Navigate to a specific **tab** from the left side panel.
- Do not copy this document to your own drive as it will be updated frequently.
- See current assignment details and upcoming assignments in your course's weekly schedule tab.

1. Overview

These courses will go back to basics, in a way. We will have a book, we will keep notes, we will have in-class discussion, and you will apply your knowledge with practical exercises.

We will, as ever, be focused on the 5 C's **and honesty**:

- Critical thinking
- Collaboration
- Communication
- Creativity
- Charisma
- Honesty

And, the Rule of Cool is always in play!

2. Grades

Work

80% of your overall grade will be earned from your completed work **and acknowledged efforts** in the form of:

- Handwritten notes (See [Notebook](#))
- Classroom activities and tasks (See [Classroom Activities](#))
- Assignments (see [LMS Assignments](#))
- Projects
- Whatever else I devise along the way
- Other **approved efforts you** devise along the way (see Rule of Cool)

This portion of the grade **WILL NOT** have a formal point system. Any points that are earned will be arbitrary and fantastical (like dragons and wizards type fantasy). **Assume you must receive a complete or acknowledged effort for all work. See [RUBRIC](#) on how work is completed or acknowledged.**

Assessments

20% will be in the form of assessments as:

- Quizzes (between 5%-10% each)
- Exams (10% max)

This portion will provide you the opportunity to earn your desired letter grade and **WILL BE** point based

Speaking of grades, my courses always use the absolute grading scale, as follows:

95%-100% → A+ (4.5 GPA)
90%-94% → A (4.0 GPA)
85%-89% → B+ (3.5 GPA)
80%-84% → B (3.0 GPA)

...

Most of you will earn one of those letter grades but everyone can earn an A+.

Everyone will pass, if you want to. If you receive an F due to attendance or lack of effort you will be given a task to receive a D for a passing grade.

All grades are negotiable!

3. LMS Assignments

Graded **assignments** will always be posted on the LMS as well as here in the individual class schedule tabs. You are required to submit for a grade. All group work must be submitted by all individual group members, even if the documents are shared. All submitted work must be created as a google drive file and submitted with its share link. Make sure file permissions are set properly.

Resubmissions are required if you make any changes to your submitted work. This will ensure I receive a notification on the LMS that indicates a change. Without a resubmission I cannot track changes accurately and may not regrade your work. Resubmissions have no limit before the due date.

What is the penalty for late or incomplete submissions? Best not to know and just complete all work.

4. Attendance

I will not use the Campus Rock attendance app to mark your attendance. I don't care if you came to class. You'll need to prove it!

University policy allows you to miss a MAXIMUM 10 days from a 3-credit course. Anything over that amount automatically triggers your failure (F).

For all my courses, you are allowed 5 days of absence penalty FREE. The next 5 absences within the allowed MAXIMUM will be penalized at -2% **of your overall grade** for each absence.

For example, if you miss 6 days you will have a maximum possible overall grade of 98% for that course. The 6th absence triggers that -2% penalty.

I prioritize **Family → Health → Nature → School → Class** in that order. What this means for you:

In order to have an absence excused you will need to provide **strong evidence** that you were engaged in one of the higher priority activities above Class overlapping our scheduled class time. Pics or it didn't happen. Let's talk about the requirements for strong proof.

5. Texts

We will use a book and I believe this is ideal for this type of class as the topics that we will cover will be few (5) but deep. The chosen book will provide the structure and cover the topics of interest for this class, and it will allow you to explore other topics that may be of interest to you.

- Just LOGISTICS:
[\(Main Text\) The Handbook of Logistics and Distribution Management](#)
[\(Supplement\) The Geography of Transport Systems](#) ← Sustainability stuff
- International TRADE:
- [\(Main Text\) International Trade: Theory and Policy \(Krugman\)](#)
[\(Secondary Text\) INTERNATIONAL TRADE THEORY AND POLICY \(Sauranovic\)](#)

- MANAGEMENT SCIENCE (**Note are not required but are suggested**):
[\(Main Text\) From theORY to Application \(Operations Research\)](#)
[\(Secondary Text\) Management Science](#)

Take a look at the book topics and do some minimal research on topics that seem of interest to you. Make a list of 5 that you would like to cover in this course.

Additional texts may be found and added to the pool of reading material during the semester. I'll be playing this as it comes to me. Supplemental texts could include:

- other books
- case studies
- academic papers
- current events
- suggested media related to your interests (with approval)

There will also be a [book club](#) this semester and ~~you must~~ I encourage you to participate.

6. Notebook

You will also keep a notebook for taking handwritten notes. This, I believe, is essential as we all do too little of this currently. Notetaking is a skill. Working the hand while reading enforces deeper connections. Notes will be written with a [specific structure](#) and you can take your own free-from notes as you like. The structure will allow me to check these for consistency and completeness. This is essential because this notebook will become the foundation for your grade in this course.

Deep reading is something I would like this course to focus on and links well with notetaking. Deep reading is the opposite of the type of "reading" you often do on the internet which takes the form of skimming and scanning, or partially reading a section of text for a "sense" of it or the gist. We are all guilty of this, unfortunately. Deep reading used to be just reading.

A daily average page goal (like steps walked per day) will be required but enforced only by you. This will ensure that you read enough of the given text to make accurate and reliable notes on the current topic. Sections of text will be given at **regular** intervals during the semester (every two weeks?). You will be responsible for completing them along with notes which will be checked. These long-form sections of text will serve as a direct contrast to the short-form you usually encounter. Ideas are often complex and CANNOT be explained simply. "Explain it like I'm a fifth grader" isn't always possible and so you'll be re-learning the skill of reading with the selected text(s) for this course. This will be painful for a portion of you but it will prove to be a valuable lesson in the end.

And, as always, your **personal, handwritten notes will be usable on all quizzes or exams.**

NOTE: If you are in multiple classes of mine, see me for instructions on notetaking.

7. Practical

A practical application of the concepts learned in this course will be required and will overlap with the management science course material. This will take the form of math, lots and lots of math. Muahaha! But, seriously, we will make use of the **computer rooms** we have been given so graciously this semester.

The practical portion of this course will use the following tools or platforms:
 Google Sheets (through google drive)

Python
AI
Github
<https://notebooklm.google/>
<https://www.ilovepdf.com/>
Etc.

We will encounter problems whose solutions may require the use of:

- Spreadsheet data / databases
- Linear Programming (LP) models
- Python programming (with AI) for creating programmatic solutions
- Web dashboards as a means of graphical and visual representation
- Decision trees
- Sensitivity analysis
- Etc.

8. Dresscode

None

9. My Past Evals

https://docs.google.com/spreadsheets/d/1tzc_c9o-ozk-zkBmOGudJK4hrMPBG6ipPQHg4S2m6q0/edit?usp=sharing

Read the comments sections for the juicy parts. I've marked my favorites. Please write another juicy one when it comes your time!

10. Extras (?)

Take a Hike!

You (or we) will.

Book Club

We will be reading 3 books over the semester, one each month. There will be a meeting at the end of each month to share opinions on the book. The next book will be decided at that time.

March - [1984](#) by George Orwell (about 30mins per day starting from 3/12 till 3/31)

April - If Anyone Builds It Everyone Dies

May - The Wizard of Earthsea

So why we do this? For fun, why else?! And the majority of you need something to do in your free time. Plus, maybe your grade will be affected...

The Sanctuary

See the [tab](#) to the left of the same name for the guided walkthrough of the task.

This is an exercise that, I believe, provides a basis for helping you find your creativity. This is a skill some people have already and one that must be gained by others. You probably all know which group you are in. This exercise will help you to place yourself.

Type Help

<https://william-rous.itch.io/type-help>

I encourage everyone to play this gem of a game. It is all text based on a simulated command line interface (CLI). This thing touched me in a way that I haven't felt in a long time.

Rules:

1. Complete the game **WITHOUT HELP** from the internet (there are full walkthroughs of course).
2. Don't use AI!
3. Keep paper notes for the beginning until the end.
4. Submit your notes to me before the end of the semester.
5. You may ask ME only for guidance.

Bike Trip (4-Rivers Cross Country)

<https://www.bike.go.kr/eng/index.do>

Complete the trip across Korea in either direction and you basically get an A+ (seem unfair? Then bring your arguments to class). You will be awarded 40% over any of the courses you have with me. The 40% may be distributed between multiple courses. % may **NOT** be applied to attendance, however the days you are absent while on the trip will not count towards the [current attendance policy](#).

Rules:

1. Plan your trip.
2. Talk to me before you go. Several times.
3. You must purchase the [stamp book](#) from a certification center ([nearest one](#)) and complete the stamps on the trail.
4. You must take selfies at all (99%) stamp booths.
5. Be safe.

This will probably be the best thing you do while in Korea. It is a challenge and a significant achievement on a few levels.

This will cost you money! I am guessing around the 500k won range, so you must do your own research. If you need specific help on that end please ask me. You may also use the bike I completed the trip with.

*A Significant Use of a C

If you can provide an example of a significant use of any ONE of the 5 Cs then you automatically receive an A+ and you no longer have to come to class (1 per course!).

The catch is, I cannot tell you what that is. If you have an idea then tell me privately. If it is indeed significant then we will both know before you finish speaking.

???

This will be revealed at the end! But, whatever you do, do not get discouraged or quit. Everything works out in the end.

1984 - Meeting Recap

Attendeez:

Asadbek

Abdumajid

Zukhriddin

Abbos

Wan xingyao

Li bingxuan

Next book suggestions:

Fahrenheit 451

Animal Farm

If they build it, everyone dies

Wizard of Earthsea

The Dispossessed

Questions asked:

What's in room 101?

Which group are we in, Proles, Outer Party, or Inner Party?

Why won't the Proles ever rise up against the Party?

Who is your "O'Brien"?

How are social media and the internet a double edged sword? How does it compare to the Party's surveillance, censorship, and fabrication of information?

Nepal's recent "revolution" as an example.

April's Book - If Anyone Builds It, Everyone Dies

If Anyone Builds It, Everyone Dies

Maybe one person read it...

May's Book - The Wizard of Earthsea

May's book is [The Wizard of Earthsea](#)

Download the epub [here](#).

It is an excellent example of world building by the beloved Ursula K. Le Guin. This book has a special connection to degrowth, the antihero, and systems balance. All things that could connect back to any of our course, even if it is a stretch.



Notes Structure

What You Need

A paper note book with pages large enough to contain the Notes Structure (see section), preferably with a spiral binding so your note book lays flat when opened.

A hard wood pencil of excellent quality. I recommend Tombow Mono J or Mono 100.

A pencil sharpener of excellent quality. Again, I'd go Japanese on this one.

Consider the following first

What is the purpose of taking handwritten notes?

What are good notes?

Why am I taking these specific notes?

Why do I need a wooden pencil?!

Notes Structure

[See Example on Serious Games](#)

[Examples of Handwritten Notes](#)

Pages 3 →

As you read, make notes from page three onward in whatever form your notes usually take. These notes should probably look unorganized and mostly intelligible to you. Use whatever language you think in. I will not read these but I will enjoy looking through your window.

First Two Pages:

- Create the structure for these first two pages before beginning your notetaking.
- Write these in English.
- Do not attempt these pages before creating your unorganized notes starting page 3.

Page one Cornell notes method

What are [Cornell notes](#)?

Split the page into two sections vertically

Left-hand narrow column includes:

- Guiding questions (3ish) (given our created together)
- Answers to guiding questions (do after the reading and after the right side is finished)

- Key words

Right-hand side:

- Main concepts
- Concept explanations (in your own words)
 - Diagrams
 - Models
 - Worked examples
 - Clarifications from discussion
 - Summary

Page two split page note method

Split the page into three sections horizontally.

Provide simple and clean examples and answers for each section.

Top section applies concepts to the real world:

- Where does this concept apply?
- What kind of business/system problem uses this?
- Who uses this?

Mid section examines the surface level application of concepts:

- Where/when would Concept A fail?
- What are the limitations of Concept B?

Bottom section reflects and asks questions still unclear:

- What wasn't addressed?
- What are you confused about?
- What ifs
- What abouts
- **What is most likely to appear on an assessment???**



Logistics Schedule

Logistics Weekly Schedule

Week 1 (March 4) -

Monday:

1. Holiday (will become a scheduled makeup day)

Wednesday:

1. Meet & Greet
-

Week 2 (March 9 & 11) -

Monday:

1. Meet & Greet 2
2. Class policies introduction

Wednesday:

1. (ALL CLASS OVERLAP) Class policies review
 2. (ALL CLASS OVERLAP) Notes structure introduction
-

Week 3 (March 16 & 18) -

Monday:

1. (ALL CLASS OVERLAP) The Sanctuary
2. (ALL CLASS OVERLAP) [Demo](#) of the note structure - pdf pages 30-33 ([Main Text](#)) [From theORY to Application \(Operations Research\)](#)
3. Introduction to the next 5 topics (working selection)
 - a. Intro to Logistics - Ch. 1, 2 (~30 pgs)
 - i. Map the logistic process
 - b. Distribution & Networks - Ch. 8, 9, 10 (~35 pgs)
 - i. Facility Location Problem
 - ii. Pareto Analysis
 - c. Inventory Management - Ch. 15, 16 or 17 (~35 pgs)
 - i. Bullwhip Effect
 - d. Warehouse Management - Ch. 18, 23 (~30 pgs)
 - i. Slotting Strategy
 - e. Transportation - Ch. 31, 32 (~30 pgs)
 - i. Traveling Salesman Problem
 - f. ???

4. [Reading 1](#): Main text Ch 1, 2 (pdf p.28-57)

- a. Notes should be on the full reading.
- b. [See example on Serious Games](#)
- c. Guiding Questions (refer to [Notes Structure](#)):

- i. Why is it essential to adopt a holistic or "Total Logistics Concept" (TLC) view of logistics rather than managing individual sub-functions like transport or warehousing in isolation?
- ii. How does globalization create a direct conflict with the goals of quick-response and just-in-time (JIT) operations?
- iii. What are the differences between logistics and supply chain?

Wednesday:

1. Assignment: Logistics Product Map

- a. Choose a product that you have recently used, then map its company's logistics chain to the source. You will conduct some basic research to complete this but, first, map out the steps in the logistics process based on your current understanding and intuition. Do this from the perspective of the product (I am candy bar). Work backward, finish to start. Do this on paper and hand it in with your name and ID.
 - b. Now, conduct a brief bit of research to determine a more accurate logistics chain for your product.
 - c. After your initial research, create a network map or flowchart of your product's logistics process. You may use any tool you like. You will submit this.
 - i. What is a network map? Flowchart?
 - ii. What tool is best?
 - d. Consider the requirements for making your own custom tool to make this process easier in the future. Imagine you are given this task at your new job. Mapping these chains for individual products can be tedious so you plan to make your own. Using the map/chart you made for your product, list the requirements needed for a software tool.
2. What to submit on the LMS:
- a. Initial logistics process on paper with name and ID (for attendance).
 - b. A flowchart or network map illustrating the logistics process for your product.
 - c. A list of requirements for your custom tool as a separate file or part of your map/chart (how will you submit this?).
 - d. Record of your process for completion.

Week 4 (March 23 & 25) -

Monday:

- 1. Housekeeping
 - a. Check notes progress
 - b. Q&A
- 2. Finish network maps / flowcharts, review issues
- 3. **Activity: Simulating Real Systems:**
 - a. Begin brainstorming for a logistics-based serious game.
 - i. Consider:
 - 1. game design
 - 2. Steps in the logistics process
 - 3. Goal of the game

4. Win requirements
5. What programming languages will you use?
 - a. Python → [jupyter notebook](#) (run python from the browser)
 - b. Javascript/HTML → host it on GitHub with Pages
6. Etc.
- ii. Get inspired:
 1. [Oregon Trail](#)
 2. [Universal Paperclips](#)
 3. [Dwarf Fortress](#)
 4. [Zork](#)
 5. [Type Help](#) (extras)
 6. [CLI Dungeon](#) (by yours truly, created for Digital Literacy to help students understand the command line interface)
 7. The Lemonade Stand (from DA)
- b. [Form your groups](#)

Wednesday

1. Activity Continued: LogSim Brainstorm

1. Create a folder to store work for your game, [here](#).
 - a. Folders can be hidden later.
 - b. Name it after your game.
 2. Then, create a shared document to store your initial brainstorming
 - a. Answer the consideration questions above.
 - b. More questions to consider
 - i. What are the referenced games about?
 - ii. What are their mechanics?
 - iii. What makes a game interesting without graphics?
 - iv. ???
-

Week 5 (March 30 & April 1) -

Monday:

1. Reading 1 due - recall session
2. Reading 1 review [presentation](#)
3. Let's talk about SMART goals again:
 - a. Specific
 - b. Measureable
 - c. Achievable
 - d. Realistic/Relevant
 - e. Time (bound)
4. What should the penalty be for late or incomplete work?
5. Catch up on LMS work
6. [Grade viewer login](#) - default password is **test**
7. **[Reading 2: Ch. 8, 9, 10 \(pdf p.146-179\)](#)**
 - a. Notes should be on the full reading.
 - b. [See example on Serious Games](#)

- c. Guiding Questions (refer to [Notes Structure](#)):
 - i. How does the "Lean" vs. "Agile" conflict drive strategic trade-offs between inventory and transport costs?
 - ii. Why must the data collection phase categorize products by characteristics like temperature or hazard to model storage costs accurately?
 - iii. How can Pareto (80/20) analysis be used to simplify the "rough-cut modelling" process in Chapter 10?
- d. One method of taking notes:
 - i. Use notebookLM with the main text and the Notes Structure from this document (2 sources total).
 - ii. Skim and scan the Reading for 4-6 topics you find interesting and important to the class. Take note from page 3 in your notebook for this reading as you skim and scan.
 - iii. Prompt NotebookLM using your chosen main concepts and the GQs (!) for a draft of the 2 page structured notes.
 - iv. Copy the draft immediately.
 - v. NOW read the Reading in full and make notes from page 3→ as you go.
 - vi. Complete a final draft of the structured notes.

Wednesday:

1. Quiz 1 ([ANSWERS](#))

a. Testable content:

- i. Readings (due before today)
- ii. This document
- iii. In class activities as documents created for participation
- iv. Any LMS submittable activities or assignments due before this date

b. Handwritten notes are allowed!

c. You will need paper and pencil

Week 6 (April 6 & 8) -

Monday:

- 1. We'll get down on some Pareto calculations and related topics.

Wednesday:

- 1. Change your file names if they do not follow the naming convention!
- 2. Understanding when to use Pareto.
- 3. **Activity: Potato Pareto Dataset:**

a. Instructions

- i. ~~Copy the data from the Dataset tab in this [sheet](#), and replace it with yours (I have changed the data).~~

- ii. Categorize the issues into one of 5 logistics steps within the TLC. Create a new column to the left of A for this.
 - iii. Understand the data with further calculations. You may need to create new columns and/or tabs for your calculations.
 - iv. Determine THREE issues to fix based on your calculations. Consider synergistic effects (+/-) of your choices!
-

Week 7 (April 13 & 15) -

Monday:

1. [Check your Pareto calculations and your three fixes.](#)
2. How are your games coming along? ([Mitigating Bananas Test](#))

Wednesday:

1. Reading 2 due - [recall session](#)
 2. Reading 2 review presentation
 3. **[Reading 3: Ch. 15, 16 \(pdf p. 226-254\)](#)**
 - a. Notes should be on the full reading.
 - b. [See example on Serious Games](#)
 - c. Guiding Questions (refer to [Notes Structure](#)):
 - i. Investigate the bullwhip effect and determine what triggers it and its consequences.
 - ii. Investigate lead-time gap. Explain the relationship between the customer and the company in these terms. Then explain the how the company addresses this gap.
-

Week 8 (April 20 & 22) - Exam Week

Monday:

1. Work on LogSim
 - a. Produce a deliverable as a functional first stage of your game.
 - b. Consider:
 - i. Product
 - ii. First stage of five
 - iii. Game mechanic(s) for the first stage
 - iv. Story
 1. 5Ws (Who, What, When, Where, Why, and sometimes How)
 2. How does this engage the player?

Wednesday:

1. **Activity: Total Logistics Cost**
 - a. Copy the sheet in [Week 8's folder](#)
 - b. Rename your file using the standard convention
 - c. Take a breath and go to your Sanctuary real quick and make sure all your Objects are still there
 - d. Complete the tasks before the end of class

Week 9 (April 27 & 29) -

Monday:

1. **Class cancelled! I'm sick enough to stay home.**

Wednesday:

1. Quiz 2 ([ANSWERS](#))
 - a. Testable content:
 - i. Completed Readings (due before today)
 - ii. This document
 - iii. In class activities as documents created for participation
 - iv. Any LMS submittable activities or assignments due before this date
2. Handwritten notes are allowed!
3. You will need paper and pencil

Week 10 (May 4 & 6) -

Monday:

1. **NO CLASS - Personal bike trip.**
2. Assignment: [May the Fourth Be with You!](#)

Wednesday:

- ~~1. Reading 3 due - recall session~~
- ~~2. Reading 3 review~~
3. How far are we on LogSim?
- ~~4. Activity: [Bullwhip & and Lead Time](#)~~
5. **[Reading 4](#): Chapters 18 & 23 (pdf p.277-284 & 346-365)**
 - a. Notes should be on the full reading..
 - b. [See example on Serious Games](#)
 - c. Guiding Questions (refer to [Notes Structure](#)):
 - i. Logistics is a set of stationary and movement systems. Which would best define warehousing? Consider this answer from the point of view of a perfectly optimized whole logistics system
 - ii. Consider the internal functions of a warehouse (some may not be defined in the reading). Which would be best suited to automation (robots and AI) and which functions would humans still be required? What conditions need to exist to change these requirements?

Week 11 (May 11 & 13) -

Monday:

1. Reading 3 due - [recall session](#)

2. Reading 3 review
3. Case study - [Watch this video that I gave you two weeks ago](#) until time 39:52

Wednesday:

1. **Activity: Presentation**
 - a. Read the Smarter Everyday Presentation [guide](#) and sign up for [your slot](#).
 2. **Activity: [Bullwhip & and Lead Time](#)**
 - a. Read the document and make a copy of the linked dataset. Use the normal naming convention when saving to the Week 10 folder.
-

Week 12 (May 18 & 20) -

Monday:

- ~~1. Presentation 1~~
2. **Continue Activity: [Bullwhip & and Lead Time](#)**
 - a. **Understanding complex questions**
 - i. **Defining terms**

Wednesday:

1. Reading 4 due - recall session
 2. Reading 4 review
 3. **Continue Activity: [Bullwhip & and Lead Time](#)**
 - a. Discuss Bullwhip effect
 - b. Calculate the rolling average to predict more accurate order numbers
 - c. Use standard deviation to help the prediction
 - d. What to do:
 - i. Complete the simple questions on your own
 - ii. Compare with a peer's answers
 - iii. Correct your answers
 4. **Activity: [Heat map](#)**
 - a. **Assess the table Question 1-3**
 5. **Reading 5**
 - a. Read both:
 - i. [Chapters 31 & 32](#) (partial) Pdf p. 504-521
 - ii. [The Forgotten Role of Logistics in Achieving Net-Zero](#)
 - b. Guiding question:
 - i. When companies improve transportation efficiency, are they making logistics truly more sustainable, or are they mainly making it cheaper and faster to move even more goods? Elaborate.
-

Week 13 (May 25 & 27) -

Monday:

1. **NO CLASS - Buddha's B-day**

Wednesday:

1. Quiz 3

a. Testable content:

- i. Readings (due before today)
- ii. This document
- iii. In class activities as documents created for participation
- iv. Any LMS submittable activities or assignments due before this date

b. Handwritten notes are allowed!

c. You will need paper and pencil

Week 14 (June 1 & 3) -

Monday:

1. Presentations
2. Watch this for Quiz 4 - <https://youtu.be/5lCWqEFVzbY?si=WoKUe1o1hvsArZHH>
3. **Continue Activity: [Heat map](#)**
4. **Activity - Slotting Strategy**

Wednesday:

1. **NO CLASS - Local election day**
 2. **HIKE OVER GUDEOK**
-

Week 15 (June 8 & 10 & 11) - Makeup Week

Monday:

1. Presentations
2. Reading 5 due - recall session
3. Reading 5 review
4. ???

Wednesday:

1. Presentations (Final day)
2. Assignment and Activity Recovery Day
 - a. Attendance is only required if you have not met the 80% minimum grade.

Thursday:

1. Assignment and Activity Recovery Day
 - a. Attendance is only required if you have not met the 80% minimum grade.
-

Week 16 (June 15 & 17) - Exam Week

Monday:

1. Quiz 4

a. Testable content:

- i. Readings (due before today)**
- ii. This document**
- iii. In class activities as documents created for participation**
- b. Any LMS submittable activities or assignments due before this date**
- c. Handwritten notes are allowed!**
- d. You will need paper and pencil**

2. Cookies

3. Notebook collection

Wednesday:

1.

Int'l Trade Schedule

International Trade Weekly Schedule

Week 1 (March 4) -

Monday:

1. Holiday (will become a scheduled makeup day)

Wednesday:

1. Meet & Greet
-

Week 2 (March 9 & 11) -

Monday:

1. Meet & Greet 2
2. Class policies introduction

Wednesday:

1. (ALL CLASS OVERLAP) Class policies review
 2. (ALL CLASS OVERLAP) Notes structure introduction
-

Week 3 (March 16 & 18) -

Monday:

1. (ALL CLASS OVERLAP) The Sanctuary
2. (ALL CLASS OVERLAP) [Demo](#) of the note structure - pdf pages 30-33 ([Main Text](#)) [From theORY to Application \(Operations Research\)](#)
3. Introduction to the next 5 topics
 - a. Introduction to Trade - Ch. 1 & 2 (~20 pgs)
 - b. Pure Exchange & Comparative Advantage - Ch. 3 & 4 (~30 pgs)
 - c. The Heckscher-Ohlin Model - Ch. 5 (~35 pgs)
 - d. Economies of Scale
 - e. ???
 - f. ???

4. [Reading 1](#) - Main text Ch. 1 & 2 (pdf p.24-46)

- a. Notes should be on the full reading.
- b. [See example on Serious Games](#)
- c. Guiding Questions (refer to [Notes Structure](#)):
 - i. What are the seven recurring themes in international economics?
 - ii. According to the gravity model, which two factors most accurately predict the volume of trade between two countries?
 - iii. How has the pattern of trade changed over the course of recorded history?

Wednesday:

1. Assignment - Ridiculous Tariffs

- a. In groups with your country-mates create a Google doc or sheet and share it between you (one person should do this). Place your first names and IDs at the top. You will submit this.
 - b. **For every one person in your group**, find a ridiculous tariff imposed by your country now OR in the past. Some countries may not have any. If there are 5 people in your group then there should be 5 different tariffs. Structure your document or sheet appropriately, in an organized and clear manner.
 - c. For each tariff, if answer the following
 - i. Why is your tariff ridiculous?
 - ii. Why does/did this tariff exist or put into place?
 - iii. What effect do you think this tariff has? Try not to look this answer up.
 - d. A brief reflection on what you learned during this activity.
-

Week 4 (March 23 & 25) -

Monday:

1. **No Class**. I have an immigration appointment but you can do the following:
2. Complete Ridiculous Tariffs (many individual group members have not completed each of the 5 parts, see above activity description).
3. Read and work on your notes.

Wednesday:

1. Activity - Autarky, AuNot?

- a. Read all instructions first!
- b. Choose a nation case from the [list](#) to research with your group (2 peeps max).
- c. With your group create a google doc / sheet in [this folder](#).
- d. Name it "CASE NAME - MEMBER_1_NAME MEMBER_2_NAME".
- e. In your shared file, complete the following together:
 - i. Define *Autarky* and *Extreme Protectionism*
 - ii. Determine if they categorize as Autarky (or near autarky) OR Extreme Protectionism. State it clearly.
 - iii. WHAT?: State the policy choice(s) in detail.
 - iv. WHY?: State the cause or reason for the policy choice(s) in detail.
 - v. WHAT?: Find evidence of black markets that arose during this period then give me the deets in brief.
 - vi. ?x2?: Answer two more interesting questions of your choice. State the question then answer it.
- f. Choose another group's case and compare then answer:
 - i. HOW?: State any fundamental differences between the WHAT and WHY of the policy choices.
 - ii. HOW?: State which case was more extreme and how.
 - iii. HOW?: State the differences or similarities between the black markets.
- g. Try to find at least one real source of each question and reference appropriately.

- h. Divide work between yourselves.
-

Week 5 (March 30 & April 1) -

Monday:

1. Reading 1 due - recall session
2. Reading 1 review [presentation](#)
3. Catch up on LMS work
4. [Grade viewer login](#) - default password is **test**
5. What should the penalty be for late or incomplete work?

6. [Reading 2](#) - Ch. 3 and 4 (partial) pdf p. 47-89

- a. Notes should be on the full reading.
- b. [See example on Serious Games](#)
- c. **Guiding Questions (refer to [Notes Structure](#)):**
 - i. What is the fundamental difference between "absolute advantage" and "comparative advantage," and which one actually determines trade patterns?
 - ii. How do we prove that a low-productivity country can still compete and benefit from trade with a high-productivity "superpower"? (I believe this was Norman's question from class)
 - iii. Interpret the general shape of the PPF model and draw a little picture for me.
- d. One method of taking notes:
 - i. Use notebookLM with the main text and the Notes Structure from this document (2 sources total).
 - ii. Skim and scan the Reading for 4-6 topics you find interesting and important to the class. Take note from page 3 in your notebook for this reading as you skim and scan.
 - iii. Prompt NotebookLM using your chosen main concepts and the GQs (!) for a draft of the 2 page structured notes.
 - iv. Copy the draft immediately.
 - v. NOW read the Reading in full and make notes from page 3→ as you go.
 - vi. Complete a final draft of the structured notes.

Wednesday:

1. Quiz 1 ([ANSWERS](#))

a. Testable content:

- i. **Readings that are due before this date.**
- ii. **This document in full (all tabs associated with this course)**
- iii. **Any LMS submittable activities or assignments due before this date**
- iv. **Activities completed class**

b. Handwritten notes are allowed!

c. You will need paper and pencil

Week 6 (April 6 & 8) -

Monday:

1. **Activity** - [Gravity Model: Understanding A](#)

Wednesday:

1. Book club from 10:15-10:45!
2. Class starts at 10:30 for everyone else. Continue working on Activity - [Gravity Model: Understanding A](#)

3. Show results of Activity - [Gravity Model: Understanding A](#)

4. Interpret A...

5. **Assignment - I Require a Story**

- a. Title: you choose
- b. Point of view: you choose
- c. Genre: scifi, historical fiction, whatever
- d. Consider the 5Ws: who, what, when, where, why
- e. Topics up to this point:
 - i. Globalization - ideas: expanding external pressure, change global for the purposes of your story
 - ii. Tariffs - ideas: conflict between two parties, tit-for-tat (DA peeps, remember what happens there?!)
 - iii. Autarky and Extreme Protectionist state (or civilization) - ideas: an extreme state or civ or situation creating utopia/dystopia (1984, yo!), look at adjectives around these topics (ex. isolated)
 - iv. Gravity model - ideas: poor predictions lead to poor outcomes, misplaced trust, poor advisor to the king
 - v. A - ideas: realization of weakness leads to a solution, $A \rightarrow 0$?
- f. RULEZ:
 - i. DONT USE AI
 - ii. It must be a cohesive story **integrating** as many topics covered thus far to **drive the plot**
 - iii. Must be fiction. Elements of non-fiction allowed.
 - iv. Must be handwritten
 - v. Can be in any language (I'll translate it if need be)
- g. Submission reqs:
 - i. Submit here as pictures of your handwritten story
 - ii. or hand it in to me before the due date.

Week 7 (April 13 & 15) -

Monday:

1. Quick look at Gravity Model answers.
2. **Activity** - [Comparative advantage](#)

Wednesday:

1. Reading 2 due - [recall session](#)
 2. Reading 2 review presentation
 3. Continue comparative advantage activity.
 4. **[Reading 3](#) - Ch. 5 (pdf p. 110-141)**
 - a. Notes should be on the full reading.
 - b. [See example on Serious Games](#)
 - c. **Guiding Questions (refer to [Notes Structure](#)):**
 - i. How does the heckscher-ohlin model differ from the Ricardian model in terms of construction and predictive potential?
 - ii. Explain factor price equalization and why specific factors might not equalize in reality.
-

Week 8 (April 20 & 22) - Exam Week

Monday:

1. Study for your exams!
2. Formative assessment on PPF (papers collected in class)
3. **Assignment:**
 - a. **Beat the AI debator!**
 - b. <https://chatgpt.com/g/g-69e55aad8f18819188593bd158bdaa52-dah-snarky>
 - i. Click the link above to start a chat with DAH-Snarky, the Devil's Advocate for Autarky!
 - ii. **Your Role:** You will learn both sides of this argument as you converse with Snarky, and in turn, you will be better able to argue your side AGAINST self sufficiency.
 - iii. **Your Goal:** Convince DAH-Snarky that Autarky under any definition is not ideal.
 - iv. **Win Condition:** Use one specific case or scenario that causes DAH-Snarky to cede the point.
 - c. **Submit on the LMS as:**
 - i. **The share link to your full chat with DAH-Snarky showing a convinced DAH-Snarky OR your eventual break down and admission that self sufficiency is best.**
 - d. [System Prompt for DAH-Snarky](#)
 - e. [My conversation](#)
 - f. [Winning arguments](#)

Wednesday:

1. **Activity - Magic Box**
 - a. Play all stages of [Magic Box](#)
 - b. Answer the questions and let me check your score
 2. Take notes on the activity
-

Week 9 (April 27 & 29) -

Monday:

1. **Class cancelled! I'm sick enough to stay home.**

Wednesday:

1. **Quiz 2 ([ANSWERS](#))**
 - a. **Testable content:**
 - i. **Completed readings (due before today)**
 - ii. **This document**
 - iii. **In class activities as documents created for participation**
 - iv. **Any LMS submittable activities or assignments due before this date**
-

Week 10 (May 4 & 6) -

Monday:

1. **No Class. Personal bike trip.**
2. **Assignment: [May the Fourth Be with You!](#)**

Wednesday:

1. **[Reading 4](#) - Chapter 7 Economies of Scale (pdf p.174-192)**
 - a. Notes should be on the full reading..
 - b. [See example on Serious Games](#)
 - c. Guiding Questions (refer to [Notes Structure](#)):
 - i. An easy one for ya: Why might countries trade even when they do not have obvious differences in resources, labor, or technology?
 - ii. When external economies of scale exist, does international trade always benefit every country involved? Why might the answer be more complicated than in earlier trade models? Think about this one.
-

Week 11 (May 11 & 13) -

Monday:

1. Reading 3 due - [recall session](#)
2. Reading 3 review
3. Policy change - all LMS assignments will be due 6 days by midnight from the date given. The next day (7) will be given for corrections and resubmissions.
4. Field trip announcement
5. Case study - [Watch this video that I gave you two weeks ago](#) until time 39:52

Wednesday:

1. Cookies
2. **Activity** - Read the Smarter Everyday Presentation [guide](#) and sign up for your slot.
3. **Activity** - [Abundant Factors](#)
 - a. Copy the sheet and rename it with the standard convention.

Week 12 (May 18 & 20) -

Monday:

1. Review quiz 2
2. **Activity:** [Systems thinking notes](#)
 - a. Complete in your notebook:
 - i. Define system
 - ii. Provide an example
 - iii. Define systems thinking

Wednesday:

1. Presentation 1 & 2
- ~~2. Reading 4 due – recall session~~
- ~~3. Reading 4 review~~
4. **Continue Activity: Systems thinking notes**
 - a. Factory example for internal econ of scale.
 - b. Can it continue forever? Why?

Friday:

5. **Reading 5 - Concept of Degrowth**
 - a. Read all three of the following in this order:
 - i. [The Carrier Bag Theory of Fiction - Le Guin](#)
 - ii. [What does degrowth mean?](#)
 - iii. [Global Value Chain Development Report](#) (pdf p. 246-252)
 - b. Guiding questions
 - i. There are none this time. Instead, consider that each of these texts tells you a story.
 1. Le Guin tells us how we might imagine our values differently
 2. Hinkle defines degrowth and what it argues
 3. GVC report how the current system is adapting to “green growth”
 - ii. Then think about: Can the existing global trade system become sustainable, or does sustainability require a deeper change in values and scale?

Week 13 (May 25 & 27) -

Monday:

1. **NO CLASS - Buddha's B-day**

Wednesday:

1. **Quiz 3**
 - a. Testable content:
 - i. Readings (due before today)

- ii. This document
 - iii. In class activities as documents created for participation
 - iv. Any LMS submittable activities or assignments due before this date
- b. Handwritten notes are allowed!
- c. You will need paper and pencil
-

Week 14 (June 1 & 3) -

Monday:

- 1. Presentations
- 2. Quiz replacement task announcement
- 3. ~~Reading 5 - recall~~
- 4. Activity - ???

Wednesday:

- 1. **NO CLASS - Local election day**
 - 2. **HIKE OVER GUDEOK**
-

Week 15 (June 8 & 10 & 11) - Makeup Week

Monday:

- 1. Presentations
- 2. Reading 5 - recall

Wednesday:

- 1. Presentations (Final day!)
- 2. Assignment and Activity Recovery Day
 - a. Attendance is only required if you have not met the 80% minimum

Thursday:

- 1. Assignment and Activity Recovery Day
 - a. Attendance is only required if you have not met the 80% minimum
-

Week 16 (June 15 & 17) - Exam Week

Monday:

- 1. **Quiz 4**

Wednesday:

MS Schedule

Management Science Weekly Schedule

Week 1 (March 4) -

Monday:

1. Holiday (will become a scheduled makeup day)

Wednesday:

1. Meet & Greet
-

Week 2 (March 9 & 11) -

Monday:

1. Meet & Greet 2
2. Class policies introduction

Wednesday:

1. (ALL CLASS OVERLAP) Class policies review
 2. (ALL CLASS OVERLAP) Notes structure introduction
-

Week 3 (March 16 & 18) -

Monday:

1. Introduction to the next 5 topics
 - a. Coming soon!
2. A math problem!
3. **Reading 1: Main text Ch. 4 (pdf p.42-67)**
 - a. Notes are not required but usable for quizzes.

Wednesday:

1. **Activity: Thinking Is Hard**
 - a. Problem 1: Dong A wants to improve student satisfaction while reducing costs. What would they do?
 - i. What is satisfaction?
 - ii. How do we measure it?
 - iii. What is the objective(s)?
 - iv. What are the constraints (limitations)?
 - b. Problem 2: Allocating Resources
 - i. You have:
 1. 40 labor hours
 2. 30 machine hours

- 3. Can produce Product A or B
 - ii. A uses more labor, but less machine hours
 - iii. B uses more machine hours, but less labor
 - iv. A has a higher profit, but variable demand
 - v. B has a lower profit, but stable demand
-

Week 4 (March 23 & 25) -

Monday:

- 1. Housekeeping
 - a. Class starts at 12:15, ends at 1:30 or before.
- 2. **Activity: The Pizza Optimizer: A Decision Problem**
 - a. Copy [this document](#) to [this folder](#)
 - b. Rename your file to "Pizza Optimizer - YOURNAME ID"
 - c. Begin activity

Wednesday:

- 1. Partial solution to $a+b+c=?$
 - a. Why did I ask you to solve this problem (how does it relate to what we will be doing)?
 - 2. Intro to Mathematical Modeling
 - a. Complete the Pizza Optimizer
 - i. From part 5, define each part with notation using Ch.4 as a reference
 - 3. **Assignment: Custom Optimization Problem**
 - a. Demo:
 - i. Download the pizza optimizer document as a markdown file or a pdf.
 - ii. Upload that as context to GPT.
 - iii. Then I would have given it a prompt like this:

```
I am a student studying linear optimization for a university Management Science course. Create five similar optimization problems to the Pizza Optimizer in the form of a word problem with the required data. Make sure they are different in structure and output from the Pizza Optimizer example problem and from each problem you create. I will be using these to practice linear optimization. Do not provide the solution! You may provide step-by-step instructions.
```
 - b. LMS submission requirements:
 - i. Your selected problem, data, and your solution in a single google sheet.
 - ii. Include the linked AI conversation or source to the problem you selected.
 - iii. Include the notation for each part of the problem (objective function, parameters, decision variables, sets, and constraints) using the standard notation given in Chapter 4.
-

Week 5 (March 30 & April 1) -

Monday:

1. ~~Reading 1 review presentation~~
2. Your inevitable questions for the Assignment
3. [Grade viewer login](#) - default password is **test**
4. What should the penalty be for late or incomplete work?
5. [A story](#)
6. **Reading 2: [Secondary Text](#) Ch. 2 (pdf p.17-46) OR/AND [Main Text](#) Ch. 5 (pdf p.68-78)**
 - a. Notes are not required but are suggested for the quizzes.
 - b. One method of taking notes:
 - i. Use notebookLM with the main text and the Notes Structure from this document (2 sources total).
 - ii. Skim and scan the Reading for 4-6 topics you find interesting and important to the class. Take note from page 3 in your notebook for this reading as you skim and scan.
 - iii. Prompt NotebookLM using your chosen main concepts and the GQs (!) for a draft of the 2 page structured notes.
 - iv. Copy the draft immediately.
 - v. NOW read the Reading in full and make notes from page 3→ as you go.
 - vi. Complete a final draft of the structured notes.

Wednesday:

1. **Quiz 1 ([ANSWERS](#))**
 - a. **Testable content:**
 - i. **Readings**
 - ii. **This document**
 - iii. **Any LMS submittable activities or assignments due before this date**
 - b. **Handwritten notes are allowed!**
 - c. **You will need paper and pencil**
-

Week 6 (April 6 & 8) -

Monday:

1. ~~Time to start! We begin optimizing. Demonstration(s) of choosing problems and solving them (analog to the assignment).~~

Wednesday:

1. Shown example of the Assignment - Pizza Optimization Problem.(binary DVs)
2. **Activity: Pinocchio Problem**
 - a. Work through a maximization problem with integer DVs
 - b. Problem Statement:

Pinocchio, Inc. manufactures 2 types of wooden toys: trucks and trains. The price of a piece of truck is 550 CZK, of a piece of train 700 CZK. The wood cost for the truck is 50 CZK, whereas for the train 70 CZK. The truck requires 1 hour of carpentry labor and 1 hour of finishing labor (assembling and painting). The train requires 2 hours of

carpentry labor and 1 hour of finishing labor. Worth of carpentry labor is 30 CZK per hour, worth of finishing labor is 20 CZK per hour. Each month, Pinocchio has 5000 available hours of carpentry labor and 3000 hours of finishing labor. Demand for trains is unlimited, but at most 2000 trucks are, at an average, bought each month. The Pinocchio's management wants to maximize monthly profit (total revenue - total cost).

3. MOVE TO MONDAY → ~~Attempt to build a general LP solver for maximization problems with continuous variables from scratch with AI!~~

Week 7 (April 13 & 15) -

Monday:

1. Finish [Pinocchio problem](#) from book.
2. **Activity: General linear equation solver creation**
 - a. Understanding the requirements
 - b. Build your prompt
 - c. Generate with AI
 - d. Test
 - e. Host on github pages
 - i. GitHub account creation necessary ([see video](#))

Wednesday:

1. **Activity: Decision Trees Notes**

Week 8 (April 20 & 22) - Exam Week

Monday:

1. Brief discussion on what to prepare for Quiz 2

Wednesday:

1. **Activity: Test your ability**
 - a. **Problem:**

A street food company has a budget of 360,000 to invest in two different types of trucks. Trucks of the first type serve tacos. They cost 30,000 each, there is a maximum of 8 available for purchase, and it is expected that will attract 2,000 customers each per week. Trucks of the second type serve burritos. They cost 40,000 each, there is a maximum of 6 available for purchase, and it is expected that will attract 5,000 customers each per week. Our goal is to determine how many trucks of the first type (decision variable x_1) and of the second type (decision variable x_2) to purchase to maximize the number of expected customers while satisfying constraints on availability and budget.
 - b. Determine the objective function and the constraints of the problem.

- c. Graph the problem to determine the feasible region and the corner point(s) as potentially optimal solutions.
 - d. Find the slope of the OF and determine the optimal solution by solving for x_2 .
 - e. [Play the Burrito optimization game](#)
-

Week 9 (April 27 & 29) -

Monday:

1. Class cancelled! I'm sick enough to stay home.

Wednesday:

1. Quiz 2 ([ANSWERS](#))
 - a. Testable content:
 - i. Completed readings (due before today)
 - ii. This document
 - iii. In class activities as documents created for participation
 - iv. Any LMS submittable activities or assignments due before this date
 2. Reading 3 - [Main Text](#) Ch. 7 Branch and Bound (pdf p.126-154)
-

Week 10 (May 4 & 6) -

Monday:

1. No Class.
2. Assignment: [May the Fourth Be with You!](#)

Wednesday:

1. Activity: [Decision tree activity](#)
 - a. <https://excalidraw.com/>
 2. ~~Present challenge quiz material~~
-

Week 11 (May 11 & 13) -

Monday:

1. Present challenge quiz on Chapter 3.
2. Activity: **Simplified Pizza Optimizer (3 toppings) solution using BB**

Wednesday:

1. Take challenge quiz
2. Try the [BB visualizer](#) (updated to view as a webpage) for a simpler version of Pizza Optimizer.

Week 12 (May 18 & 20) -

Monday:

1. Review quiz 2
2. **Activity: [Networks - TSP](#)**
 - a. What is the lowest cost with all available nodes?
 - b. Are there more paths with the same cost?
 - c. How can we find all possible paths of the lowest cost?

Wednesday:

1. **Continue Activity: [Networks - TSP](#)**
 - a. What is the lowest cost with all available nodes?
 - b. Are there more paths with the same cost?
 - c. How can we find all possible paths of the lowest cost?
2. **Assignment: Branch and Bound**
 - a. **Requirements:**
 - i. From 5/6, your completed group Decision Tree (DT) from one of the 3 scenarios.
 - ii. The decision tree of your completed Pizza Optimizer problem solution from the spreadsheet you created way back for this problem.
 - iii. And the implemented BB technique of the same problem using continuous (decimal) decision variables instead of binary ($\{0,1\}$).
 - b. Note: 2 and 3 are different but will look similar. Refer to the main text for Branch and Bound and my [visualizer](#) for the simplified problem seen in class.
 - c. All requirements can be provided as neatly hand drawn diagrams in pencil, an excalidraw file **for all three**, or a program (html or python) solution with visualization.

Week 13 (May 25 & 27) -

Monday:

1. **Red day - Buddha's B-day**

Wednesday:

1. **Quiz 3**
 - a. **Testable content:**
 - i. **Readings (due before today)**
 - ii. **This document**
 - iii. **In class activities as documents created for participation**
 - iv. **Any LMS submittable activities or assignments due before this date**
 - b. **Handwritten notes are allowed!**
 - c. **You will need paper and pencil**
-

Week 14 (June 1 & 3) -

Monday:

1. Watch Khan's [video on Branch and Bound](#)
2. Presentations
3. Watch this for Quiz 4 - <https://youtu.be/kS-CGkiPetQ?si=GvPARcioXvZYIRpy>
4. **Activity - Picking Heuristic**
 - a. Scenario: You are a warehouse picker and you have 12 items to pick as quickly as possible. Each item is located in a random location around the warehouse. What is the best method to collect all 12 items?
 - b. Requirements:
 - i. Draw a warehouse map and place your 12 items. Use this as a visual reference.
 - ii. Create three heuristics to produce a "good" collection time. Do not use nearest neighbor.

Wednesday:

1. **NO CLASS - Local election day**
 2. **HIKE OVER GUDEOK**
-

Week 15 (June 8 & 10 & 11) - Makeup Week

Monday:

1. **Continue Activity - Picking Heuristic**
 - a. Compare picking heuristics

Wednesday:

1. Assignment and Activity Recovery Day
 - a. Attendance is only required if you have not met the 80% minimum

Thursday:

1. Assignment and Activity Recovery Day
 - a. Attendance is only required if you have not met the 80% minimum
-

Week 16 (June 15 & 17) - Exam Week

Monday:

1. Quiz 4

Wednesday:

- 1.



The Sanctuary

The Sanctuary

The following idea came about from a [book](#) I came upon recently. You will not need paper, pencil, or any other materials for this exercise. Not now, not later.

You will be creating a place of solitude using only your imagination called your Sanctuary. Yes, I'm serious. Once created you will place at least 10 Objects inside your Sanctuary.

To begin, imagine you are in a wide open Space outside. A place within nature. A pleasant place. It could be a field of grass with rolling hills in the distance. Maybe a clearing in a forest you once visited as a child. Or even a wide sunny beach on a deserted island surrounded by an endless ocean of rolling waves.

Place yourself there with the sun high above you. Your shadow is the only thing that accompanies you. You and your shadow stand in the blank slate where you will build a shelter. Can you see where you are? If not, do not continue until you are satisfied with your Space.

Now, pick a spot and conjure your shelter from the emptiness around you. This will become your Sanctuary. It may be of any size, shape, or complexity; a small single room cottage, a castle with a hundred rooms, a treehouse in the only tree nearby, or even a tethered hot air balloon.

Start to imagine now the details of your Sanctuary as you view it from the outside. What is it made of? Which direction does the entrance face? If you turn in that direction what do you see? How do you enter your Sanctuary?

But, before you enter, you must enter alone. You must leave your shadow behind. Create an anchor for your shadow. Stand next to your anchor. Reach down and pluck your shadow from your feet and attach it to this anchor. Say goodbye to your shadow for the time being. Now you are ready to enter your Sanctuary.

Try to feel the weight of your door as you open it, the texture of the rope ladder to your treehouse, or the damp walls that lead you down into the earth. However you enter, imagine the feel of it..

Once you are inside your Sanctuary you can begin to see the details more clearly, because most likely before this moment you have already decided on a few things. Those details will shift and solidify eventually. The layout and things that populate the space or room you have just entered will slowly take form. Once you can see it clearly, then you may begin to create your Objects.

You need to decide what these Objects are and where they will go. Standing in your Sanctuary you will conjure these Objects into being and place them where they belong. It may be best to let the first thing that comes to your mind be the current Object, even if you don't like what it is. You will find a purpose for it later. Do this for each Object and before you are done you may realize you have more than 10. You will not forget them.

Step back from your creations, once finished, and observe them where you put them. Try to hone in on the details of each Object. Now observe your sanctuary as a whole. Turn towards the exit and walk out. Retrieve your shadow and come back to this reality.

This process must be completed at each step to be successful. Once finished, you will find that you want to go back to your Sanctuary to explore, to look at your Objects, to remember them, to detail them further, to build your world.

What does the Space represent?

What does your shadow represent?

What is the purpose of the Objects?

How does this link to note taking?

Which of the 5 Cs does this link to and how?

What does your sanctuary say about you?

What is world building and why have we always done it?